Smart Campus Assistant

Software Requirements Specification (SRS)

Smart Campus Assistant	Project Name
1.0	Version
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8.3.26	Date

1. Project Overview

1.1 Purpose of the System

The **Smart Campus Assistant** is an AI-based system designed to help students quickly access campus-related information.

Students can ask questions in natural language and receive answers about various campus services, including:

- exam schedules
- office hours
- classroom and building locations

The system uses an **AI service** to analyze the user's question and generate a response based on information stored in a campus database.

The system also includes an **Admin interface** that allows administrators to add, update, and delete information stored in the database.

The system includes secure administrator authentication.

Only authenticated administrators can access the admin panel and modify campus data.

The system also includes fallback mechanisms in case the external AI service is unavailable.

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1.2 System Goals

The main goals of the system are:

- Provide quick access to campus information
- Reduce repetitive questions directed to university staff
- Allow administrators to easily maintain campus data
- Use AI to generate clear and natural responses for users

Role and Interest	Stakeholder
Main users who want quick access to campus information	Students
Responsible for maintaining and updating the information stored in the system	admin
Build and maintain the system	Developers
External system used to analyze questions and generate answers	AI Service
Stores all campus information used by the system	Database
Responsible for maintaining the system and monitoring failures	IT Support

2. Use Cases

The system supports several main use cases.

UC-01 – Ask a Campus Question

Actor: Student

Goal:
Get information about campus services.

Precondition:
The student is using the system interface.

Main Flow

1. The student enters a question.
2. The system sends the question to the backend.
3. The backend sends the question to the AI service for analysis.
4. The system retrieves relevant information from the database.
5. The AI service generates a response.
6. The system displays the answer to the student.

Postcondition

The student receives the requested information.

UC-02 – View Exam Schedule

Actor: Student

Goal:
Receive information about exam dates and times.

Precondition:
The student is using the system interface.

Main Flow

1. The student enters a question asking about the schedule of a specific exam.
2. The system sends the question from the user interface to the backend server.
3. The backend sends the question to the AI service to determine that the question is related to an **exam schedule**.
4. The backend queries the database for the requested exam information.
5. The database returns the relevant exam details (course name, exam date, exam time, and room if available).
6. The backend sends the retrieved exam information together with the original question to the AI service.
7. The AI service generates a clear natural language response containing the exam details.
8. The backend returns the generated response to the user interface.
9. The system displays the exam schedule information to the student.

Postcondition

The student receives the requested exam schedule information.

UC-03 – Find Room or Building Location

Actor: Student

Goal:
Find the location of a classroom or building on campus.

Precondition:
The student is using the system interface.

Main Flow

1. The student enters a question asking for the location of a room or building.
2. The system sends the question to the backend server.
3. The backend sends the question to the AI service to determine that it is a **location-related query**.
4. The backend queries the database for the requested room or building information.
5. The database returns the location details (building name, room number, floor, etc.).
6. The backend sends the question and the retrieved information to the AI service.
7. The AI service generates a natural language response containing the location information.
8. The system displays the answer to the student.

Postcondition:
The student receives the location of the requested room or building.

UC-04 – Add Campus Information

Actor: Admin

Goal:
Add new campus information to the database.

Precondition:
The admin is authenticated and has access to the admin panel.

Main Flow

1. The admin opens the admin panel.
2. The admin enters new information.
3. The system saves the information to the database.

Postcondition:
The new campus information is saved in the database and becomes available to the system.

UC-05 – Update Campus Information

Actor: Admin

Goal:
Update existing information in the database.

Precondition:
The admin is authenticated and has access to the admin panel.

Main Flow

1. The admin selects the existing campus information that needs to be updated.
2. The admin edits the relevant fields and submits the changes.
3. The system updates the information in the database and confirms the update.

Postcondition:
The selected campus information is updated in the system.

UC-06 – Delete Campus Information

Actor: Admin

Goal:
Remove outdated or incorrect information from the system.

Precondition:
The admin is authenticated and has access to the admin panel.

Main Flow

1. The admin selects the campus information that should be removed.
2. The admin confirms the deletion action.
3. The system deletes the selected record from the database.

Postcondition:
The outdated or incorrect information is removed from the system.

Edge Cases

Edge Case A – Question cannot be classified

If the AI service cannot classify the user's question,
the system displays an error message asking the user to rephrase the question.

Edge Case B – No data found

If the database query returns no results,
the system informs the user that the requested information is not available.

Edge Case C – AI Service Unavailable

If the external AI service is unavailable or returns an error,

the system will return a fallback message informing the user that the AI service

is temporarily unavailable.

If relevant information was already retrieved from the database,

the system may return a simplified response based only on the database data.

3.Functional Requirements

Requirement	ID
The system shall allow students to submit questions through the user interface	FR1
The system shall send the question to the AI service for analysis	FR2
The system shall retrieve relevant information from the database	FR3
The system shall generate a natural language response	FR4
The system shall display the generated answer to the student	FR5
The system shall allow admins to add new campus information	FR6
The system shall allow admins to update existing information	FR7
The system shall allow admins to delete outdated information	FR8
The system shall authenticate administrators before allowing access to the admin panel.	FR9
The system shall restrict database modification actions to authenticated administrators only.	FR10

The system shall return a fallback response if the external AI service is unavailable.	FR11

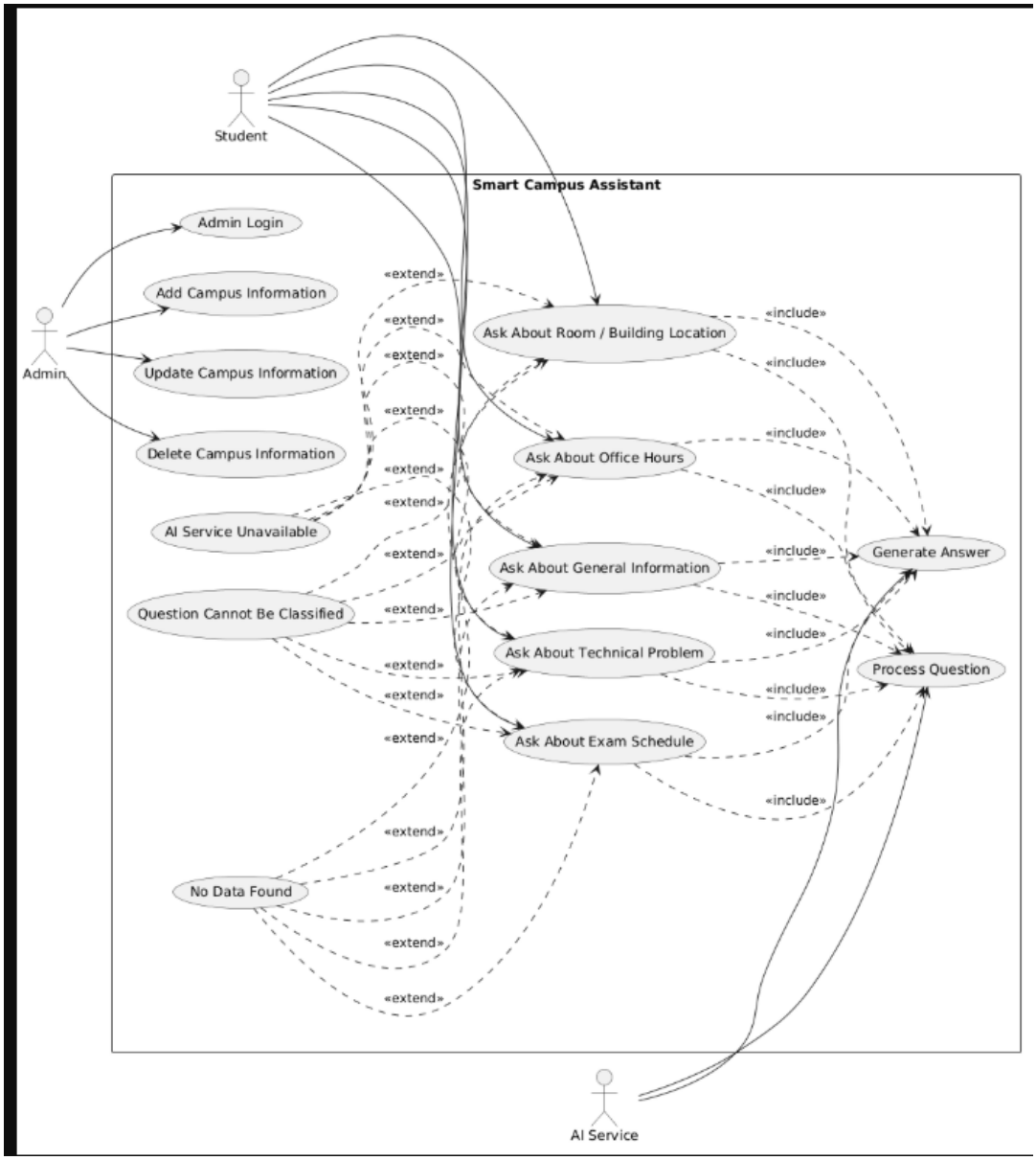
4. Non Functional Requirments

Measurable Standard	Requirement	Category
The system should return answers within 3 seconds	Response time	Performance
The system should be available at least 95% of the time	Availability	Reliability
Admin authentication will be implemented using JWT-based authentication. Only authenticated administrators with valid tokens can access admin endpoints and modify database information.	Admin access	Security

The interface should be simple and intuitive	Interface	Usability
The system should support multiple students simultaneously	Concurrent users	Scalability

5. Use Case Diagram

The diagram shows the main actions that users can perform in the system and the actors involved.



The Use Case Diagram illustrates the main interactions between the system actors and the Smart Campus Assistant system.

Students can ask different types of campus-related questions such as exam schedules, room locations, office hours, general information, and technical issues. These requests are processed by the system, which uses an AI service to analyze the question and generate a response based on information stored in the database.

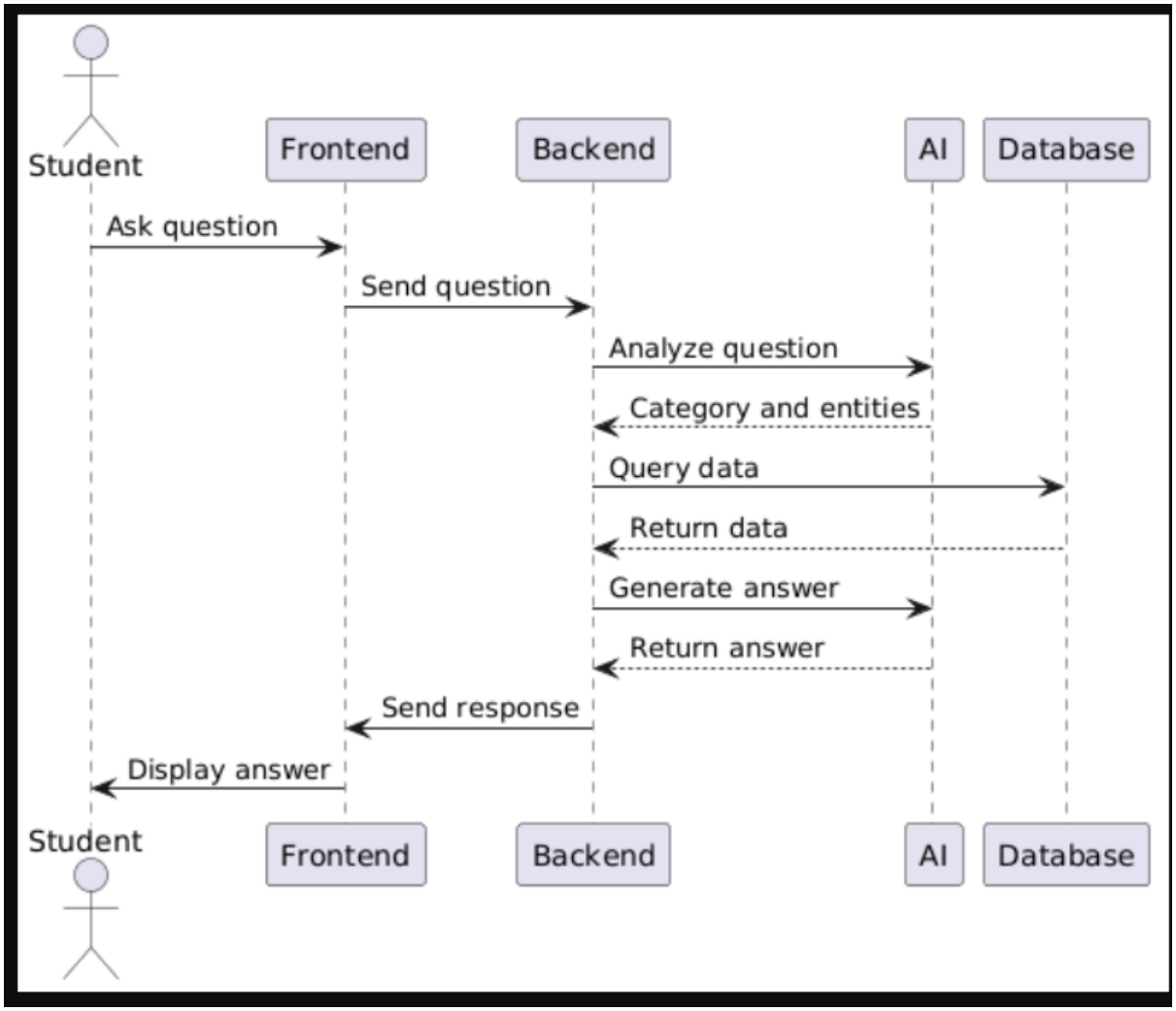
Administrators are responsible for managing campus data through the admin panel, including adding, updating, and deleting information.

Edge cases such as unclassified questions, missing data, and AI service failures are also represented in the diagram.

6. Sequence Diagram

The sequence diagram shows the order of interactions between the system components when a student asks a question.

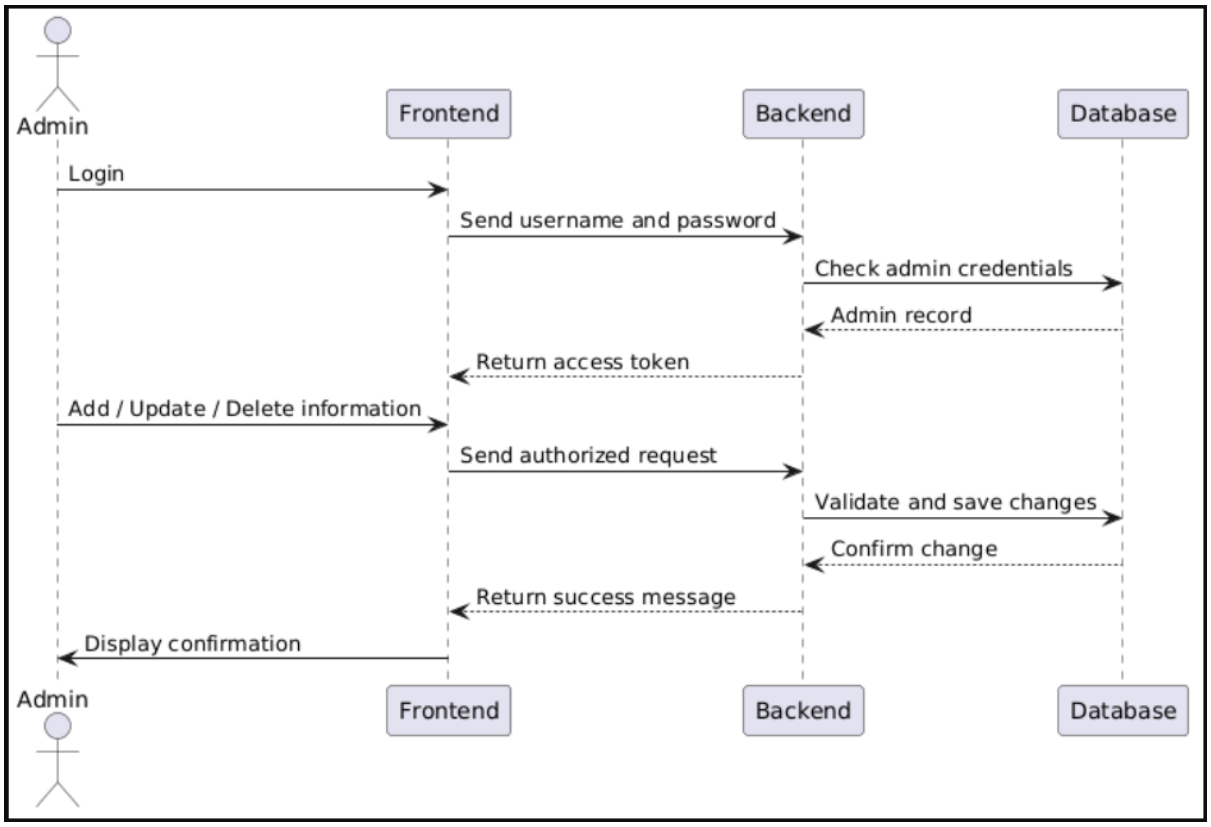
Student sequence diagram



To reduce response time and API costs, the system combines question classification and entity extraction into a single AI request.

Admin sequence diagram

The sequence diagram shows the order of interactions between the system components when a admin make changes.



The admin sequence diagram describes how an administrator securely logs into the system and performs database management actions. After authentication, the admin can send authorized requests to add, update, or delete campus information. The backend validates the request, updates the database, and returns a confirmation message to the admin.